K. VIJAY PRASANTH

ASPIRING SOFTWARE ENGINEER • BACKEND & CLOUD INFRASTRUCTURE

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CAREER OBJECTIVE

Motivated final-year Computer Science and Engineering student with strong foundational knowledge in backend development, database management, and cloud architecture. Seeking an entry-level software engineering role to leverage skills in Java, Spring Boot, and enterprise infrastructure to build scalable, high-performance solutions.

EDUCATION

B.E. in Computer Science and Engineering **2022 – 2026**

Mahendra College of Engineering, Salem | ***CGPA: 7.14 / 10.0***

TECHNICAL SKILLS

Programming Languages:	Java, SQL
Frameworks & Tools:	Spring Boot, RESTful APIs, Git, GitHub, Postman, Linux CLI
Databases:	MySQL, PostgreSQL (Relational Modeling, Query Optimization, ACID Properties)
Core Competencies:	Database Management, Cloud Infrastructure, Enterprise Virtualization, Network Security & Cryptography, Object-Oriented Software Engineering (OOSE)

ACADEMIC PROJECTS

Industrial Pollution Monitoring System **Java, Spring Boot, SQL, REST APIs**

- Designed and implemented an automated monitoring framework to capture, evaluate, and report environmental emission metrics in real time.
- Structured normalized relational backend data schemas and high-throughput SQL database queries for continuous tracking and reporting.
- Engineered robust RESTful API endpoints using Java and Spring Boot to streamline telemetry ingestion and dispatch automated threshold breach alerts.

Healthcare Assistant / Medical Chatbot **Java, NLP Logic, SQL, Database Management**

- Developed an interactive conversational assistant to facilitate immediate symptom inquiry and deliver automated, accurate healthcare guidance.
- Integrated natural language parsing logic with structured backend data retrieval mechanisms to ensure reliable response mapping.
- Designed modular architecture and query routines to retrieve diagnostic references and clinical support data efficiently.

CERTIFICATIONS & KEY COURSEWORK

- **Cloud Service Management & Infrastructure:** Cloud deployment architectures, hypervisor operations, enterprise virtualization, and resource provisioning.
- **Cryptography & Network Security:** Ciphers, public-key infrastructure (PKI), data integrity protocols, authentication mechanisms, and network security.
- **Object-Oriented Software Engineering (OOSE):** Software development lifecycles (SDLC), UML modeling, architectural patterns, and maintainable design.